

LYMPHOID NEOPLASIA

PKMYT1 is a targetable vulnerability in del(17p) high-risk multiple myeloma

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KEY POINTS

- PKMYT1 is highly expressed in del(17p) MM, and its inhibition represents a vulnerability.
- PKMYT1 inhibitor triggers DNA damage and mitotic catastrophe, leading to apoptosis in del(17p) MM cells.

Deletion of chromosome 17p [del(17p)] is among the most adverse cytogenetic abnormalities in multiple myeloma (MM). By integrating RNA sequencing data from cells of patients with MM with genetic dependency data from MM cell lines, we identified protein kinase membrane-associated tyrosine/threonine 1 (PKMYT1), a member of the WEE family, as a potential therapeutic target in MM cells harboring del(17p). Genetic suppression or pharmacological inhibition of PKMYT1 activity with the selective inhibitor RP-6306 triggered accumulation of DNA damage, micronucleus formation, and mitotic catastrophe, resulting in preferential cell death in del(17p) MM cells while largely sparing del(17p)⁻ MM cells and healthy cells. RP-6306 also reduced tumor burden and extended survival in vivo in both xenograft and TP53-deficient syngeneic models. Collectively, our findings nominate PKMYT1 as an actionable target and support PKMYT1 inhibition as a biomarker-driven therapeutic strategy for patients with del(17p) or TP53-deficient MM.

Introduction

Multiple myeloma (MM) is a malignant disorder characterized by the clonal proliferation of plasma cells (PCs) in the bone marrow (BM) that remains largely incurable. Although treatment advances have greatly extended the overall survival (OS) of most patients, those carrying the high-risk genetic abnormality deletion of chromosome 17p [del(17p); ~10% of newly diagnosed patients] have not benefited as much from these advancements, even in the era of quadruplet therapies.¹⁻⁴ In contrast to solid tumors, where both TP53 alleles are often knocked out, most patients with del(17p) still retain 1 functional copy of TP53, yet these patients still have a bad prognosis.^{5,6} In fact, accumulating evidence confirms the independent prognostic impact of del(17p) on both OS and progression-free survival.⁷ Despite the need for therapies to specifically treat this patient subset, few studies in MM and other malignancies have attempted to target specific vulnerabilities engendered by the loss of 17p. Some studies have found that the heterozygous deletion of TP53 often encompasses a neighboring essential gene, POLR2A, which renders cancer cells susceptible to further inhibition of POLR2A.^{8,9} In vitro, a B-cell maturation antigen antibody conjugated to a POLR2A inhibitor has superior activity in del(17p) myeloma cells.¹⁰

To find more such vulnerabilities, we performed a comprehensive RNA sequencing (RNA-seq) analysis to identify global gene expression changes in patients with del(17p) compared with control MM and normal PCs. We identified protein kinase membrane-associated tyrosine/threonine 1 (PKMYT1; also known as MYT1) as upregulated in del(17p) MM cells and a predictor of poor outcome. Moreover, by interrogating genetic dependency data from MM cell lines, we observed an enhanced genetic dependency on PKMYT1 in del(17p) MM cells.

PKMYT1 is a member of the WEE family of serine/threonine kinases that regulates the G2/mitosis (G2/M) phase transition by inhibiting cyclin-dependent kinase 1 (CDK1) activity, ensuring genome integrity.^{11,12} Compared with WEE1 kinase (WEE1), PKMYT1 is not essential for cell cycle progression.^{6,13} However, its overexpression or hyperactivation is linked to poor prognosis in several cancers,¹⁴⁻¹⁶ and its inhibition has shown synthetic lethal activity with CCNE1 amplification,¹⁷ and FBXW7 or PPP2R1A loss of function mutations.^{18,19}

RP-6306, also known as lunresertib, is a first-in-class oral small-molecule inhibitor of PKMYT1.^{16,20} Here, we show that treatment with the PKMYT1 inhibitor RP-6306 specifically targeted

del(17p) MM cell lines and primary cells from patients with del(17p), leading to irreparable DNA damage and cell death both in vitro and in vivo. This study may provide the rationale for developing personalized medicine in del(17p) MM.

Methods

RNA-seq data sets of patients with MM

BM samples were collected from 192 patients newly diagnosed with MM (NDMM) (Toulouse data set), plus 319 patients with NDMM and 16 healthy donors (IFM2009 data set). All patients signed an informed consent form, allowing research on their samples, and the study was accepted by the Toulouse Ethics Committee. The independent patient RNA-seq data set contains 507 samples from patients with NDMM and 21 normal PCs. Patients in the Toulouse data set were randomly selected. A collaboration request for the data set can be submitted via myelomaresearch.net.

Cell lines

All myeloma cell lines were cultured in RPMI-1640 (Gibco) supplemented with 10% fetal bovine serum (Gibco, Life Technologies, Carlsbad, CA), 1% penicillin/streptomycin, and 1% L-glutamine, with 2.5 ng/mL interleukin-6 for INA6, ANBL6, and XG1 cells. Primary MM cells and BM mononuclear cells were isolated using Ficoll-Hypaque density gradient sedimentation and CD138 microbead separation from BM aspirates of patients with MM after informed consent was provided and after Dana-Farber Cancer Institute Institutional Review Board approval. The murine primary MM cell lines MM5080 and MM7027 have been previously described.²¹ These cells harbor heterozygous deletions of the *Trp53* gene. As a control, a *Trp53* wild-type (WT) murine primary MM cell line was used. Cells were maintained in appropriate culture medium supplemented with 10% fetal bovine serum, penicillin G (100 U/mL), and streptomycin (0.1 mg/mL) and cultured at 37°C in a humidified atmosphere containing 5% CO₂. Cultures were routinely tested and confirmed to be mycoplasma-free.

Reagents

RP-6306 (lunresertib; a PKMYT1 inhibitor), RO-3306 (a CDK1 inhibitor), MG132, melphalan, and bendamustine were purchased from Selleck. Compounds were dissolved in dimethyl sulfoxide.

Generation of CRISPR/Cas9-KO cell line

Cells were spin-infected with an all-in-one lentiviral vector delivering both single-guide RNA (sgRNA) and Cas9 nuclease, purchased from Horizon Discovery (Lafayette, CO). The all-in-one CRISPR-PKMYT1 knockout (KO) lentiviral sgRNAs included clone IDs VSGHSOMG_30423180, VSGHSOMG_30380280, and VSGHSOMG_30470430, catalog no. GSGH12181-15EG9088, and nontargeting control lentiviral sgRNAs included clone ID SVC19050802 catalog no. GSGC12058. Positive cells were sorted by flow based on their green fluorescent protein (GFP) expression.

Cell viability

Cell viability was measured by CellTiter-Glo Luminescent Cell Viability Assay (Promega; catalog no. G7572), and luminescence was measured using the SpectraMax M Series

(Molecular Devices). Fifty percent inhibitory concentration (IC₅₀) values for each human myeloma cell line were determined using nonlinear regression analysis in GraphPad Prism.

Colony formation assay

Five hundred H929 or KMS11 cells were resuspended in 2 mL Human Methylcellulose Complete Media Without Erythropoietin (Epo) (R&D Systems; catalog no. HSC004) and seeded in a 24-well plate. Vehicle or 0.5 μM RP-6306 was added to the final volume of 2 mL. The experiment was run in triplicate. After 14 days, wells were stained with 300 μL 1 mg/mL nitroblue tetrazolium chloride (Millipore Sigma; catalog no. N6876), and 24 hours later, colonies were counted using the GelCount colony counter (Oxford Optronix).

BrdU, apoptosis, and DNA damage staining

Cells were seeded in T-25 flasks and treated with RP-6306 for 24 hours. Bromodeoxyuridine (BrdU) was added to the cultures for 1 hour at the end of the treatment (at a final concentration of 10 μM). Cells were collected and processed with the Apoptosis, DNA Damage and Cell Proliferation Kit (BD Biosciences; catalog no. 562253) following the manufacturer's protocol. Samples were acquired using a Fortessa flow cytometer (BD Biosciences), and the data were analyzed with FlowJo version 10.10.0 software.

Annexin V and propidium iodide staining

Cells were seeded in T-25 flasks and collected after 96 hours of treatment with RP-6306. Cells were resuspended in 1× Annexin-V binding buffer (BD Biosciences; catalog no. 556454) and stained with 2 μL of Annexin-V fluorescein isothiocyanate (BioLegend; catalog no. 640906) for 15 minutes at room temperature. Then, 1 μL of propidium iodide (BD Biosciences; catalog no. 51-6621E) was added to each condition. Samples were acquired using a Fortessa flow cytometer (BD Biosciences), and the data were analyzed with FlowJo version 10.10.0 software.

Micronucleus assay

The cells were cultured in the presence or absence of RP-6306 and were evaluated for micronuclei, a marker of genomic instability, using the Micronucleus Assay MicroFlow Kit (Litron Laboratories, New York, NY).

Humanized MM model in severe combined immunodeficiency mice

All mouse experiments were approved by and conducted according to the relevant regulatory standards of the Institutional Animal Care and Use Committee at the Dana-Farber Cancer Institute. Six-week-old male NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ mice were purchased from The Jackson Laboratory (stock number 005557). Mice were maintained in pathogen-free conditions, housed in ventilated cages, and fed ad libitum. A total of 50 × 10³ cells/μL KMS11 cells were subcutaneously injected into the mice. Upon detection of measurable tumors (~100 mm³ tumor volume), mice were randomly assigned to different experimental groups: (1) vehicle, (2) low-dose RP-6306 (5 mg/kg), and (3) high-dose RP-6306 (10 mg/kg). Although undergoing treatment, mice were routinely weighed. The tumor measurements were carried out using a caliper, and tumor volumes (cubic millimeters) were

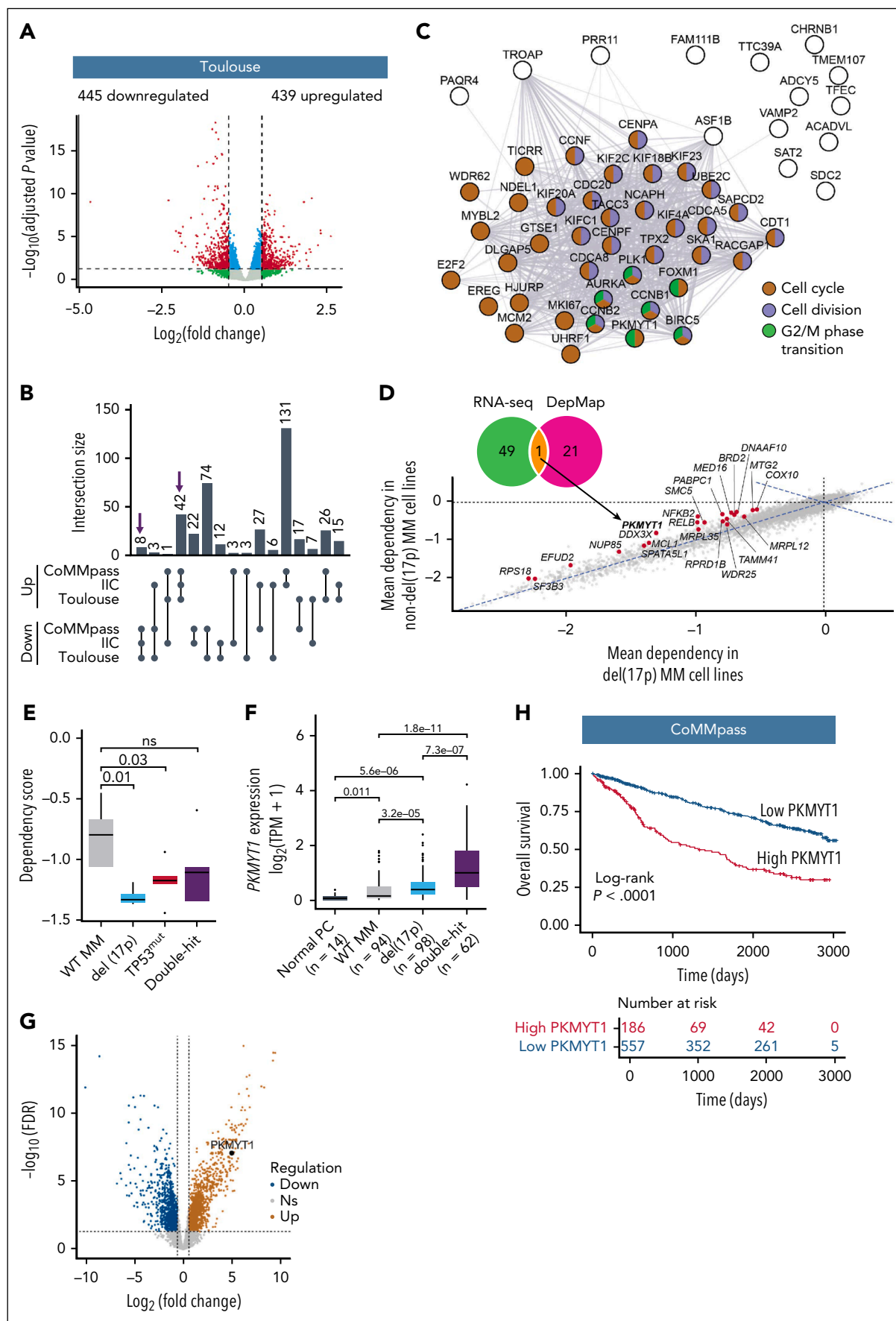


Figure 1.

calculated as follows: (length × width²)/2. The end point was established at a tumor volume higher than 2000 mm³, and mice were then euthanized with CO₂. Tumors were removed, dissociated using gentleMACS Dissociator (Miltenyi Biotec; catalog no. 130-093-235), and incubated with trypsin-EDTA (0.05%; Gibco; catalog no. 25300054) for 10 minutes at 37°C under rotation. Tumor cells were then filtered and pelleted. OS and progression-free survival curves were estimated using the Kaplan-Meier method and were compared using the log-rank test.

MM5080-derived syngeneic model

Syngeneic transplants were established by injection of 5 × 10⁶ MM5080 cells resuspended in Dulbecco phosphate-buffered saline into the tail vein of 8- to 10-week-old C57BL/6 mice of both sexes. Mice (n = 8) were randomly assigned to 3 treatment groups: (1) vehicle, (2) RP-6306 at 5 mg/kg, or (3) RP-6306 at 10 mg/kg. Treatment was initiated on day +3 after injection. RP-6306 was prepared in 10% dimethyl sulfoxide and 90% of a 20% sulfobutyl ether β-cyclodextrin solution in saline and administered orally twice daily, 5 days per week, for 3 weeks. Therapeutic responses were evaluated using Kaplan-Meier survival analysis and compared using the log-rank test. All in vivo experiments were approved by the University of Navarra Ethics Committee for Animal Experimentation.

Statistical analysis

The significance of differences between experimental variables was analyzed using the statistical test specified in each legend. Analyses were performed using either R version 4.3.2 or GraphPad Prism version 10.

Results

PKMYT1 is a functionally and clinically relevant target in del(17p) MM cells

To investigate therapeutically actionable vulnerabilities in del(17p) MM, we integrated genome-scale dependency screens in MM cell lines with gene expression profiles in MM cells from patients. We first analyzed RNA-seq data from the CD138⁺ MM cells of 98 patients with NDMM harboring

del(17p) and 94 patients with non-del(17p)/TP53 WT (supplemental Table 1, available on the *Blood* website). Relative to standard risk MM, del(17p) cases exhibited extensive differential expression, with 445 genes downregulated and 439 upregulated (Figure 1A); notably, 17% of the downregulated genes (75/445) reside on 17p. Integrating these results with 2 additional MM cohorts (Multiple Myeloma Research Foundation CoMMpass study and an internal data set) yielded 50 genes recurrently dysregulated in del(17p) MM (Figure 1B). Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) analysis indicated that a large proportion of these genes are involved in cell cycle regulation and cell division, with a subset enriched for G2/M phase transition (Figure 1C; supplemental Figure 1A).

By leveraging data from genome-wide Cas9-based CRISPR KO screens, we identified 22 genes with a greater effect on MM cell lines with del(17p) compared with MM cell lines without del(17p) (Figure 1D). Combining these data with the transcriptomic profile of MM cells from patients with del(17p) MM, we identified the PKMYT1 kinase as the only dysregulated gene with a greater dependency in del(17p) MM cell lines compared with those without del(17p) (WT) (mean dependency score, −1.30 in del(17p) vs −0.89 for WT cells; *P* = .01) (Figure 1D-E; supplemental Figure 1B). We further confirmed a trend toward higher dependency in del(17p) cells in 2 additional genome-wide CRISPR-KO data sets in MM cell lines (supplemental Figure 1C).

In patients, PKMYT1 expression increased from normal PC to MM cells without del(17p) and was significantly upregulated in del(17p) and double-hit MM cells (*P* < .0001) (Figure 1F). Using single-cell transcriptomics on patients with subclonal del(17p), we confirmed within the same patient that PKMYT1 was overexpressed in tumor cells harboring del(17p) compared with cells without del(17p) (Figure 1G; supplemental Figure 1D), and it was within the top 30 upregulated genes in the 5 patients examined. Finally, high expression of PKMYT1 was associated with poor OS (hazard ratio, 3.1; *P* < .0001) in the CoMMpass data set (Figure 1H).

Figure 1. Integrated transcriptomic and functional analyses identify PKMYT1 as a target in del(17p) MM. (A) Gene expression profile showing differentially expressed genes between monoallelic del(17p) (n = 98) and non-del(17p) MM (WT) (n = 94). The vertical dashed lines are at log₂ fold change of 0.5 and −0.5, and the horizontal dashed line represents the threshold for −log₁₀ adjusted *P* value of .05. (B) Dysregulated genes across the Toulouse and CoMMpass cohorts, plus an independent internal cohort (IIC) with 507 samples from patients with NDMM. Fifty genes were commonly dysregulated: 42 upregulated genes (*P* = 9.99 × 10^{−5}) and 8 downregulated genes (*P* = .0005). The significance of the number of overlapping genes was determined by a bootstrap analysis. (C) Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) analysis showing the existence of a highly interconnected functional protein network among the 50 commonly dysregulated genes in del(17p) vs non-del(17p). *MIR22HG* (lncRNA) was excluded by STRING. (D) Mean dependency scores between del(17p) (n = 4) and non-del(17p) (n = 7) MM cell lines based on Dependency Map (DepMap) Chronos dependency score (22Q4+Score database). The 22 genes with a more significant dependency in del(17p) MM cell lines are highlighted in red. These genes were selected as follows: genes with a significant statistical difference between del(17p) and non-del(17p) cell lines using a Mann-Whitney test, genes with a mean dependency score below −0.5 in del(17p) cell lines, and genes with a difference between the mean dependency in non-del(17p) and the mean dependency in del(17p) below 0. The Venn diagram shows the overlap between the 50 commonly dysregulated genes in del(17p) patients with NDMM and the 22 genes with a significant preferential dependency in del(17p) MM cell lines; only PKMYT1 was found to be a common gene. (E) PKMYT1 Chronos scores stratified by 17p status in MM cell lines [double-hit defined as concurrent del(17p) and TP53 mutation]. Gene dependency data (Chronos scores) were obtained from the Cancer DepMap Broad Institute project (DepMap Public Release 23Q4; <https://depmap.org/portal/>). Non-del(17p) MM (WT), n = 5; del(17p), n = 4; TP53-mutated, n = 5; double-hit abnormality, n = 5. Data were analyzed using the global Kruskal-Wallis test for global comparison (*P* = .044) and the Mann-Whitney test for pairwise comparisons. (F) PKMYT1 transcripts per million (TPM) expression in PC among healthy subjects (normal PC) and patients with NDMM stratified by their del(17p) status in the Toulouse data set. Patients with double-hit abnormality (n = 62) were also included in this analysis. Data were analyzed by the global Kruskal-Wallis test for global comparison and the Mann-Whitney test for pairwise comparisons. (G) Volcano plot showing significantly differentially expressed genes (DEGs) between cells with and without del(17p). Each point refers to a DEG. The blue and orange dots represent the downregulated DEGs (n = 872) and upregulated DEGs (n = 884) in the del(17p) cell group, respectively. The gray dots represent the nonsignificant DEGs (n = 6412). PKMYT1 is represented by the black dot. The vertical lines represent a log₂ fold change threshold of 1.5 and −1.5. The horizontal line represents the significance threshold for a −log₁₀ false discovery rate of 5%. (H) Kaplan-Meier OS of patients with NDMM from the MMRF-CoMMpass data set according to PKMYT1 TPM expression (high, top quartile; with the rest in the low group). Kaplan-Meier survival curves were compared by the log-rank test. FDR, false discovery rate; MMRF-CoMMpass, Multiple Myeloma Research Foundation CoMMpass study; ns, not significant.

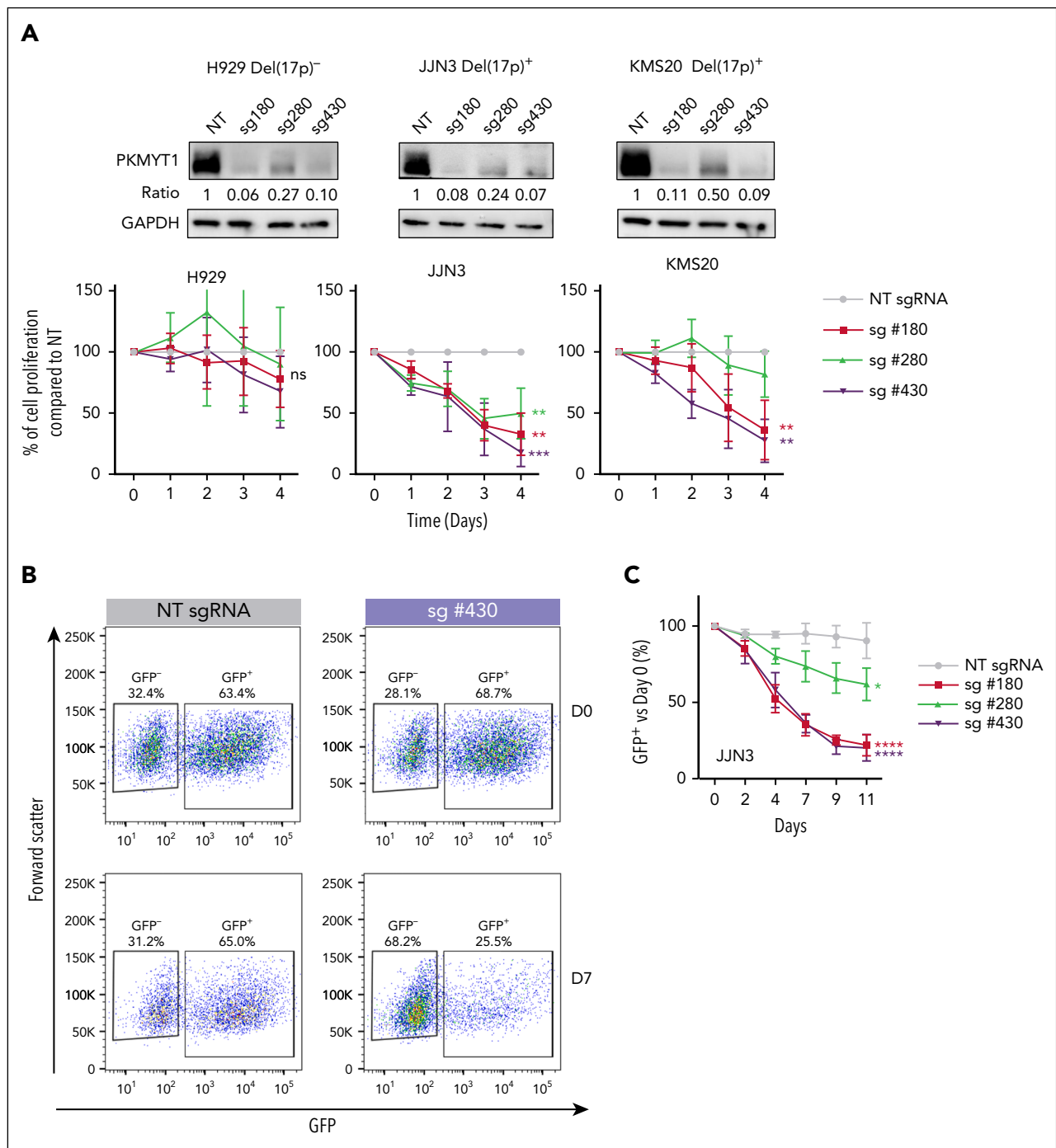


Figure 2. Genetic depletion of PKMYT1 represses del(17p) MM cell viability. (A) KMS20, JJN3, and H929 MM cell lines were infected with either GFP-labeled non-targeting (NT) control or all-in-1 CRISPR-PKMYT1 KO lentiviral sgRNAs and FACS sorted 72 hours after infection. Top panel: Western blot analysis was performed for evaluation of PKMYT1 expression in the H929, JJN3, and KMS20 cell lines after infection with control or PKMYT1 sgRNAs. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was used as a loading control, and the PKMYT1/GAPDH ratio is shown. Bottom panel: Cell proliferation was evaluated over time by CTG assay in non-del(17p) (H929) and del(17p) (JJN3, KMS20) MM cell lines after CRISPR-mediated PKMYT1 KO. Data show the percentage of cell growth compared with NT control cells at each given time point. Data represents mean $\pm$ SD from 3 independent experiments performed with 3 technical replicates. Statistical significance on Day 4 was determined by 1-way analysis of variance (ANOVA) with Dunnett's multiple-comparison test. (B-C) Del(17p)⁺ JJN3 cells were infected with either NT control or all-in-1 CRISPR-PKMYT1 lentiviral sgRNAs (3 replicates). Three days after infection, cells were seeded and evaluated over time by flow cytometry. Shown is the fraction of live, sgRNA-expressing (GFP⁺) cells over time compared with day 0. Representative dot plots are shown in panel B, whereas mean $\pm$ SD of 3 independent experiments performed in triplicate is shown in panel C. Statistical significance on day 11 was determined by 1-way ANOVA with Dunnett's multiple-comparison test. * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$. FACS, fluorescence-activated cell sorting; CTG, CellTiter-Glo; SD, standard deviation.

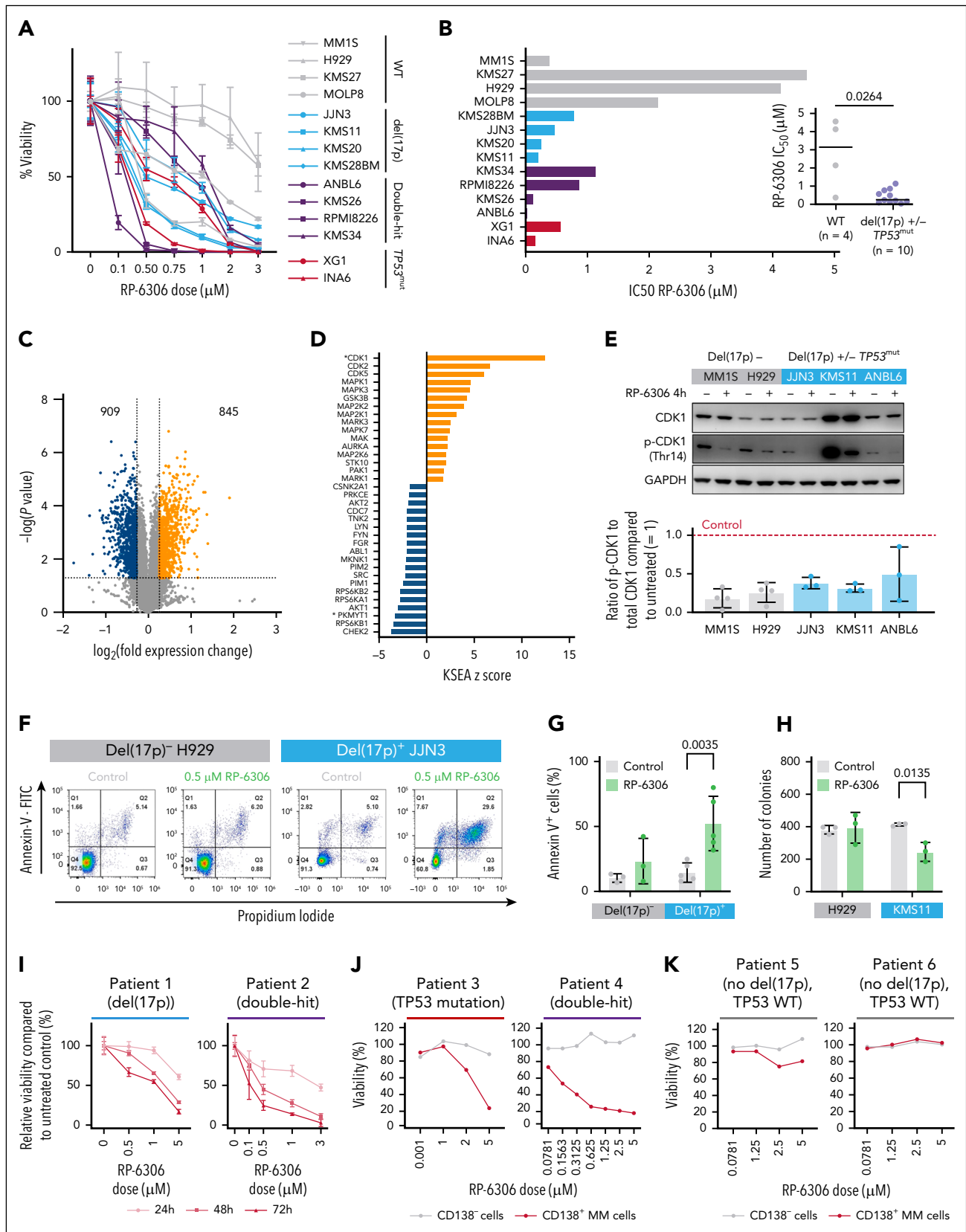


Figure 3. Pharmacological inhibition of PKMYT1 impairs viability and induces apoptosis in del(17p)- and p53-altered human myeloma cell lines (HMCLs). (A) Dose-response curves in a panel of 14 HMCLs (gray, WT; blue, del(17p); red, TP53 mutation; purple, del(17p) + TP53 mutation). Cell viability was assessed using a CellTiter-Glo (CTG) assay after 96 hours of treatment and presented as percent of cell viability compared to untreated cells. Each cell line was evaluated in at least 3 independent experiments performed in triplicate. Mean $\pm$ SD of all independent experiments is shown. (B) RP-6306 IC₅₀ was calculated using GraphPad Prism. Differences between the 2 groups of cell lines were evaluated with a 2-tailed Mann-Whitney test. (C) Tandem mass tag (TMT)-based phosphoproteomics of del(17p)+ KMS11 cells treated with 0.5 μ M

Collectively, these functional genomics and patient transcriptomic data identify PKMYT1 as a del(17p)-associated, clinically relevant dependency in MM and nominate it as a promising therapeutic target for this high-risk subgroup.

PKMYT1 is a genetic vulnerability in del(17p) MM cells

PKMYT1 expression was found to be significantly higher both at the RNA and the protein levels in del(17p) MM cell lines compared with non-del(17p) MM cell lines (supplemental Figure 2A-C). CRISPR/Cas9-mediated genetic depletion reduced PKMYT1 expression (Figure 2A; supplemental Figure 2D), leading to a time-dependent decrease in cell viability in 2 MM cell lines with del(17p); however, the same PKMYT1-targeting sgRNAs did not have a similar effect in the non-del(17p) MM cell line H929. The inhibition of cell growth was confirmed by flow cytometry analysis, where the expression of PKMYT1-targeting sgRNAs resulted in depletion of the sgRNA⁺ or GFP⁺ cell population over time (Figure 2B-C; supplemental Figure 2E).

PKMYT1 is critical in maintaining synthesis-phase and G2/M-phase DNA damage checkpoints by negatively regulating CDK1 through Thr14 and Tyr15 phosphorylation. Concordantly, we detected decreased CDK1 phosphorylation at T14 after PKMYT1 genetic depletion (supplemental Figure 2F). Among the G2 checkpoints, only the loss of PKMYT1 function was significantly more effective in del(17p) vs non-del(17p) MM cell lines (supplemental Figure 2G). Altogether, these data suggest that del(17p) may confer vulnerability to the inhibition of PKMYT1.

Pharmacological inhibition of PKMYT1 with RP-6306 impairs del(17p) MM cell growth

To evaluate the therapeutic potential of PKMYT1 inhibition in del(17p) MM cells, we used RP-6306 (lunresertib), a recently developed selective inhibitor of PKMYT1's catalytic activity,²⁰ currently in phase 1/2 clinical trials for the treatment of advanced gynecologic and digestive tumors (<https://clinicaltrials.gov/> NCT05601440, NCT05147272, NCT05605509, NCT04855656, NCT05147350, NCT06107868). Treating a panel of MM cell lines, including non-del(17p), monoallelic del(17p), TP53-mutated, and double-hit MM cell lines, with RP-6306 caused a time- and dose-dependent decrease in cell viability in MM cells harboring del(17p) and/or TP53 mutation

(Figure 3A; supplemental Figure 3A), with a significantly lower IC₅₀ in this subgroup compared with non-del(17p) MM cell lines (Figure 3B). MM1S represents an exception to the del(17p)-associated trend, as this del(17p)/TP53 WT line remains sensitive to RP-6306, suggesting that additional factors contribute to PKMYT1 inhibitor sensitivity in MM. Notably, MM1S exhibits high basal CCNE1 expression, which may underlie this response. Indeed, in our MM cell line panel, we observed a trend between RP-6306 response and CCNE1 expression, although it was not significant (supplemental Figure 3B).

Short exposure to RP-6306 (6 hours) significantly affected the phosphorylation status of numerous sites in the del(17p) KMS11 MM cell line, including reduced phosphorylation at threonine 14 (T14) and tyrosine 15 (Y15) on CDK1, consistent with PKMYT1's role as a kinase (Figure 3C; supplemental Table 2; supplemental Figure 3C). Kinase-substrate enrichment analysis was performed to identify kinases whose activity is altered by treatment with RP-6306. This analysis confirmed CDK1 kinase activity as the most positively enriched, whereas PKMYT1 kinase activity, as expected, was significantly inhibited (Figure 3D). This effect was also observed 48 hours after treatment, whereas only a limited number of proteins were affected (supplemental Figure 3D-E).

Reduced phosphorylation at threonine 14 (T14) and tyrosine 15 (Y15) on CDK1 was further investigated by western blot analysis and observed in both non-del(17p) and del(17p) cells (Figure 3E). However, RP-6306-induced CDK1 hyperactivation was only lethal to del(17p) MM cell lines (Figure 3A-B; supplemental Figure 3A), and adding the CDK1 inhibitor RO-3306 partially reversed cell growth inhibition after RP-6306 treatment (supplemental Figure 3F), suggesting the importance of CDK1 hyperactivation in the growth defect caused by PKMYT1 inhibition in del(17p) cells.

We further confirmed that initial growth inhibition was followed by irreversible apoptotic cell death in del(17p) cells (Figure 3F-G), as shown by time-course washout studies (supplemental Figure 3G). In addition, treatment with RP-6306 significantly reduced the clonogenic potential of del(17p) MM cells, whereas non-del(17p) MM cells remained unaffected (Figure 3H; supplemental Figure 3H). Finally, we confirmed the effect of PKMYT1 inhibition ex vivo in primary MM cells from patients with and without del(17p) and/or TP53 mutation. Evaluation of RP-6306 treatment in CD138⁺ cells from a patient with a

Figure 3 (continued) RP-6306 for 6 hours vs untreated control; 15 465 phosphorylation sites were identified, with significantly altered sites highlighted in the plot. A fold change >1.2 was considered upregulated, whereas a fold change <0.8 was defined as downregulated (vertical black dotted lines). Statistical significance was determined at $P < .05$ (horizontal black dotted line). (D) KSEA of TMT-based phosphoproteomics data showing significantly enriched kinases ($P < .05$). Positive scores (orange) indicate increased kinase activity, whereas negative scores (blue) indicate decreased activity. (E) Top: western blot of total CDK1 and phospho-T14 CDK1 in non-del(17p) (MM1S, H929) and del(17p) cell lines (JUN3, KMS11, and ANBL6) after a 4-hour exposure to vehicle (–) or 0.5 μM RP-6306 (+). Bottom: relative quantification of the pCDK1/CDK1 ratio normalized to GAPDH between non-del(17p) (gray) and del(17p) (blue) MM cell lines. The points inside each bar graph indicate the number of independent experiments performed for each cell line. The error bars represent mean ± SD. (F) Representative dot plots of Annexin V and PI staining in non-del(17p) MM cells (H929) and del(17p) cells (JUN3) after 96-hour treatment with vehicle or 0.5 μM RP-6306, analyzed by flow cytometry. (G) Percentage of Annexin V⁺ cells in non-del(17p) MM cells (KMS27, MM1S, and H929) and del(17p) cells (KMS11, JUN3, KMS20, ANBL6, and RPMI8226) after a 96-hour treatment with vehicle or 0.5 μM RP-6306, analyzed by flow cytometry. Two-way analysis of variance (ANOVA) with Šidák's multiple-comparison test was used. The error bars represent mean ± SD. (H) Quantification of the number of H929 or KMS11 colonies after 14 days with vehicle or 0.5 μM RP-6306 in methylcellulose semisolid medium. Graph shows mean ± SD of 3 replicates per cell line, analyzed by 2-way ANOVA with Šidák's multiple-comparison test. (I) Time- and dose-dependent effects of RP-6306 on primary MM cells from 2 patients, assessed using CTG. Patient 1 harbored del(17p) with 60% clonality; patient 2 was double-hit with del(17p) and TP53 c.396G>C (p.K132N) (97% VAF). Cell viability was measured relative to untreated controls and is presented as mean ± SD of triplicate. (J-K) ExviTech automated flow cytometry platform (Vivia Biotech, Madrid, Spain) was used to evaluate the activity of RP-6306 against primary myeloma cells in BMBCs from 4 patients with MM after 120-hour treatment with different RP-6306 concentrations. Cell survival is also shown to demonstrate the absence of toxicity of the RP-6306 inhibitor against healthy CD138⁺ population. FITC, fluorescein isothiocyanate; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; KSEA, kinase-substrate enrichment analysis; PI, propidium iodide; SD, standard deviation VAF, variant allele frequency.

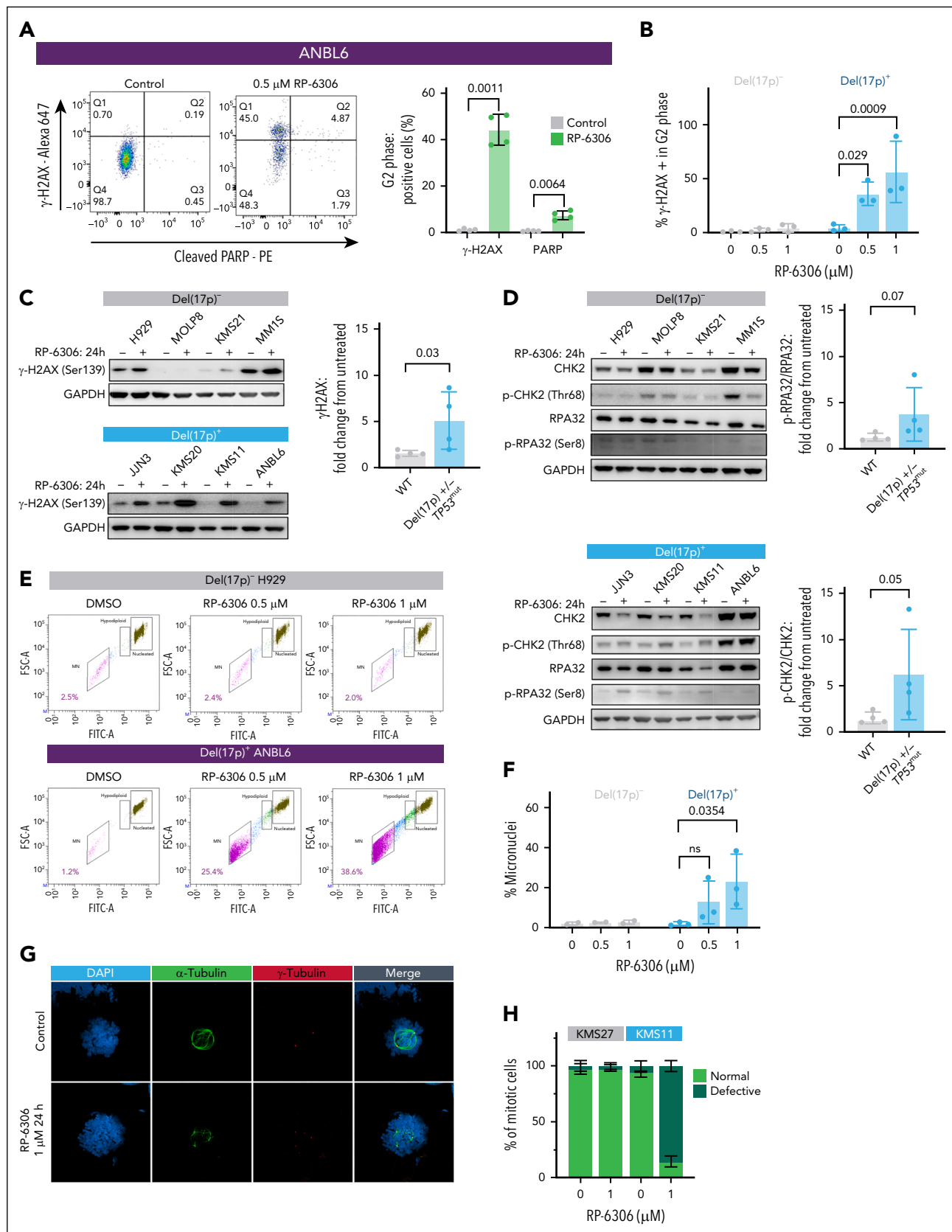


Figure 4. Abrogation of PKMYT1 activity increases DNA damage and replication stress in del(17p) MM cells. (A) Left: representative flow cytometry staining of intracellular γ H2AX and cleaved PARP in G2⁺ ANBL6 [del(17p)⁺TP53^{mutation}] cells after 24-hour treatment with vehicle or 0.5 μ M RP-6306. Right: bar plot with mean $\pm$ SD of 4 independent experiments, assessed by 2-tailed paired t tests. (B) Percentage of γ H2AX⁺ cells within G2⁺ non-del(17p) cells (H929, KMS27, and MM1S), del(17p), or double-hit cells (KMS20, KMS11, and ANBL6) after 24-hour treatment with and without RP-6306. Each cell line was tested in 3 independent experiments and then grouped based on

secondary PC leukemia (60% del(17p) clonality) and CD138⁺ cells from a patient progressing after 6 cycles of DARA-RVd induction (del(17p) and *TP53* mutation) confirmed a dose- and time-dependent activity of RP-6306 against primary MM cells (Figure 3I). Moreover, treatment of BM mononuclear cells from BM aspirates of 4 patients with MM showed a greater effect on primary MM cells harboring a *TP53* mutation or double-hit abnormality (Figure 3J) compared with non-del(17p) MM cells (Figure 3K). The IC₅₀ values determined in these bulk primary cultures are likely to be influenced by the relative abundance of del(17p) cells within each individual sample. Importantly, no significant toxicity was observed in CD138⁺ cell populations (Figure 3J-K), further supporting its selective antimyeloma effect. These data suggest a rationale for targeting PKMYT1 with RP-6306 in patients harboring del(17p) and/or *TP53* mutation.

RP-6306 induces DNA damage and replication stress in del(17p) MM cells

CDK1 controls DNA replication and the G2/M phase transition,²² and its overactivation induces replication stress through aberrant firing of replication origins.²³ Therefore, we tracked cell cycle progression and DNA replication after PKMYT1 inhibition using BrdU pulse labeling and flow cytometry, along with γ H2AX and poly(ADP-ribose) polymerase (PARP) staining to detect DNA damage and apoptotic cell death. Short-term treatment with RP-6306 caused the accumulation of γ H2AX⁺ cells in del(17p) cell lines, and the effect was more pronounced in the G2/M⁺ population (Figure 4A-B; supplemental Figure 4A-B). The induction of DNA damage was not associated with a large increase in PARP cleavage, suggesting it preceded the induction of apoptosis that was observed at a later time point (Figure 3F). We confirmed by western blot analysis that PKMYT1 inhibition by RP-6306 tended to increase the levels of γ H2AX and replication stress markers (phospho-checkpoint kinase 2 [phospho-CHK2] and phospho-RPA32) in del(17p) MM cell lines, but not in 17p-intact MM cells (Figure 4C-D). Interestingly, in the MM1S (non-del(17p)/*TP53* WT) cell line, RP-6306 induced a delayed DNA damage response with minimal changes after short exposure, as shown in Figure 3E, but clear induction after 24 hours of treatment (data not shown). *PKMYT1* KO produced comparable effects to RP-6306, with del(17p) *PKMYT1* KO cells experiencing DNA damage and replication stress (supplemental Figure 4C).

To further support the presence of heightened DNA damage in del(17p) cells after treatment, we observed a marked and statistically significant increase in micronucleus formation, an

established indicator of replication stress and genomic instability (Figure 4E-F). Finally, PKMYT1 inhibition led to the formation of supernumerary centrosomes and a disorganized spindle and induced chromosome pulverization (a marker of mitotic catastrophe) in del(17p) cells, but not in MM cells with intact 17p (Figure 4G-H; supplemental Figure 4D). Consistent with this heightened mitotic vulnerability and genomic instability, PKMYT1 inhibition also increased the sensitivity of del(17p) cells to DNA-damaging agents (supplemental Figure 4E).

RP-6306 selectively targets *TP53*-deficient MM cells, reducing tumor growth and prolonging survival in xenograft and syngeneic models of MM

To assess the effectiveness of RP-6306 against del(17p) cells *in vivo*, NOD scid gamma (NSG) mice were inoculated with KMS11 cells. Upon reaching a tumor volume of ~200 mm³, we randomly assigned the mice to 3 treatment groups: (1) vehicle, (2) low-dose RP-6306 (5 mg/kg), and (3) higher-dose RP-6306 (10 mg/kg) twice daily for 5 days per week for 3 weeks. Treatment with RP-6306 at low and higher doses significantly reduced tumor volume compared with vehicle treatment in mice (Figure 5A) and prolonged survival (Figure 5B). Monitoring mouse body weights and blood counts demonstrated that treatment with RP-6306 did not cause toxicity *in vivo* (Figure 5C; supplemental Figure 5A).

Mimicking our *in vitro* results, *in vivo* treatment with RP-6306 increased the expression of γ -H2AX and phospho-CHK2 in tumor cells retrieved from mice (Figure 5D-F; supplemental Figure 5B-D), with a trend toward lower phospho-CDK1 (Figure 5G). Notably, tumor cells harvested at end point remained sensitive to RP-6306 when tested *ex vivo* (supplemental Figure 5E).

Furthermore, the efficacy of PKMYT1 inhibition against *TP53*-altered MM cells was tested in murine MM cell lines established from primary myelomas developed in genetically engineered mice.¹⁹ RP-6306 selectively reduced the viability of 2 *Trp53*-deficient murine MM cell lines, MM5080 and MM7027, in a dose- and time-dependent manner, whereas a *Trp53* WT primary MM cell line was barely affected under identical conditions (Figure 5H). Consistent with these *in vitro* findings, RP-6306 treatment markedly prolonged the survival of immunocompetent C57BL/6 mice transplanted with MM5080 cells carrying *Trp53* loss. Thus, mice receiving RP-6306 at 5 mg/kg or 10 mg/kg twice daily showed a dose-dependent improvement in OS compared with vehicle-treated controls (Figure 5I). BM

Figure 4 (continued) the presence/absence of del(17p). Šidák's multiple-comparison test was used to calculate the *P* value. Error bars are mean $\pm$ SD. (C) Left: representative western blot analysis of 3 independent experiments of whole-cell lysates from control and treated (0.5 μ M RP-6306, 24 hours) del(17p) MM cell lines (JN3, KMS20, KMS11, and ANBL6) and non-del(17p) MM cells (H929, MOLP8, KMS21, and MM1S) using the indicated antibodies. GAPDH was used as a loading control. Right: densitometric analysis. The ratio of phospho-H2AX/GAPDH was calculated with ImageJ and is presented as fold change from untreated cells. Four cell lines for each group [non-del(17p) and del(17p)] are shown, analyzed by a 1-tailed *t* test. Error bars are mean $\pm$ SD. (D) Left: representative western blot analysis of 3 independent experiments of whole-cell lysates from control and treated (0.5 μ M RP-6306, 24 hours) del(17p) MM cell lines (JN3, KMS20, KMS11, and ANBL6) and non-del(17p) MM cells (WT) (H929, MOLP8, KMS21, and MM1S) using the indicated antibodies. GAPDH was used as a loading control. Right: densitometric analysis. The ratios of phosphorylated/total forms of CHK2 and RPA32 were analyzed with ImageJ and are presented as fold change from untreated cells. Four cell lines for each group [non-del(17p) and del(17p)] are shown, analyzed by a 1-tailed *t* test. (E) Representative plots of micronuclei quantification by flow cytometry showing dose-dependent accumulation of micronuclei after a 24-hour treatment with RP-6306 in the ANBL6 cell line (del(17p)⁺*TP53*^{mutant}), but not in the non-del(17p) MM cell line (H929). Micronuclei, pink population; nuclei, green population. (F) Micronuclei quantification (error bars, mean $\pm$ SD) after PKMYT1 inhibition for non-del(17p) (2 cell lines) and del(17p) (3 cell lines) MM cell lines, analyzed by 2-way ANOVA with Šidák's test. (G) Representative images (63 $\times$ lens objective) of mitotic KMS11 [del(17p)] cells after 24-hour treatment with 1 μ M RP-6306 or control. (H) Quantification of normal and defective mitotic cells in KMS27 and KMS11 MM cell lines (3 replicates). Error bars represent mean $\pm$ SD. ANOVA, analysis of variance; CHK2, checkpoint kinase 2; DAPI, 4',6-diamidino-2-phenylindole; DMSO, dimethyl sulfoxide; PARP, poly(ADP-ribose) polymerase; SD, standard deviation.

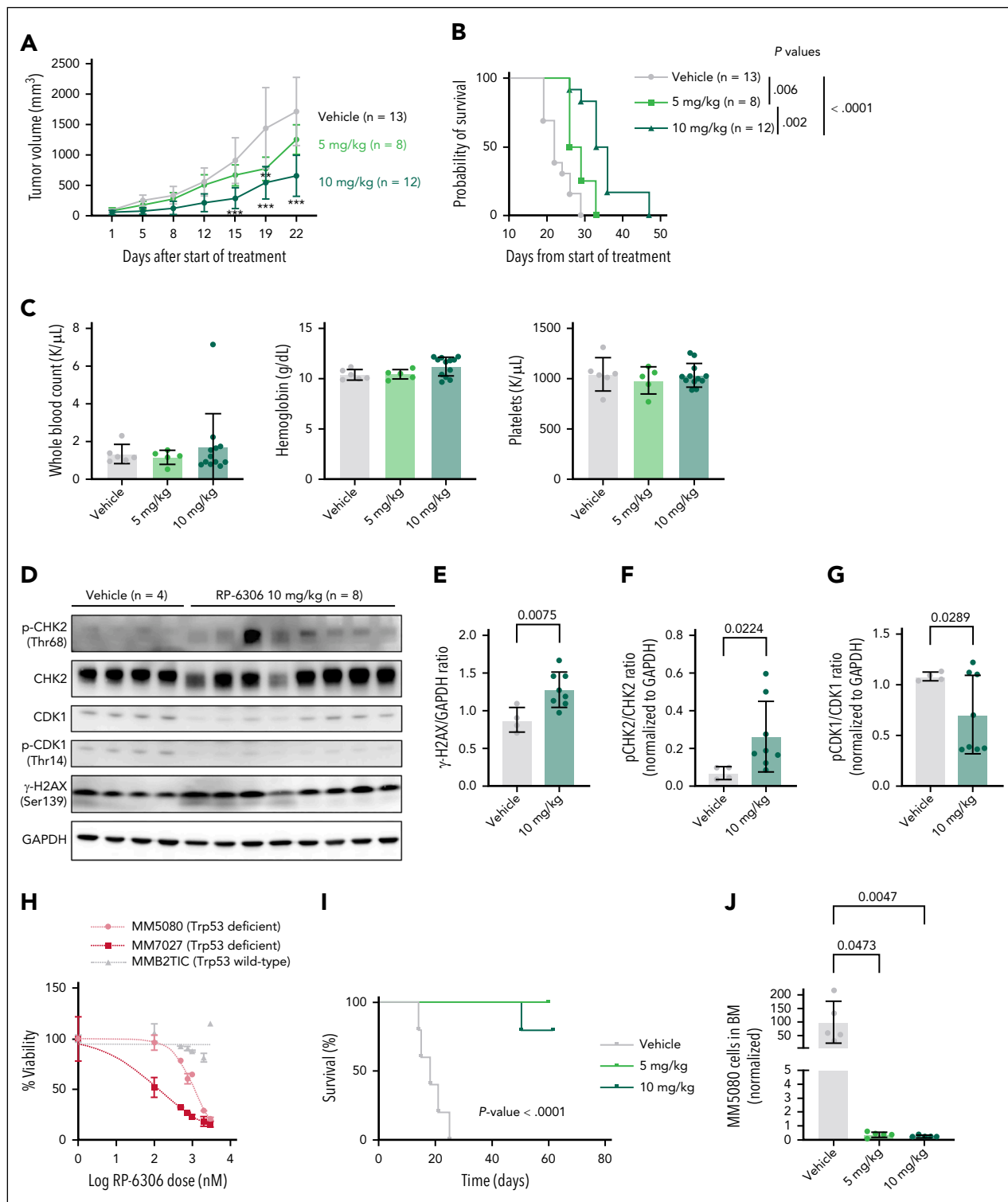


Figure 5. RP-6306 significantly reduces tumor volume and prolongs survival in mice. (A) Sublethally irradiated SCID mice were injected subcutaneously with KMS11 cells. Mice were randomized and treated with either RP-6306 or vehicle at first detection of tumor (tumor volume, ~200 mm³). Mice received either 5 or 10 mg/kg RP-6306 for 5 consecutive days per week for 3 weeks. Tumor volume was measured in 2 perpendicular dimensions by caliper twice a week. Baseline values were not significantly different among groups. Analyzed by a mixed-effects model. ***P < .01; ****P < .001. Error bars represent mean $\pm$ SD. (B) Survival was evaluated from the first day of treatment until death. Survival curves (Kaplan-Meier) were analyzed by the log-rank test. (C) Quantification of WBC, PLT, and Hb levels in mice treated with vehicle (n = 6) or 5 mg/kg (n = 5) or 10 mg/kg (n = 12) of RP-6306. Analyzed using unpaired t tests. Error bars represent mean $\pm$ SD. (D) Western blot analysis was performed on lysates from tumors excised from representative mice and probed with the indicated antibodies. (E-G) Protein band intensities were quantified using ImageJ. (E) γ -H2AX/GAPDH ratio was calculated for each tumor. The mean γ -H2AX/GAPDH ratio for each group is shown, expressed relative to untreated cells. (F-G) Signals were first normalized to GAPDH, and data are presented as the mean ratio of phosphorylated to total protein for each group. Statistical analysis was performed using unpaired t test with Welch's correction. Error bars represent mean $\pm$ SD. (H) Primary murine MM cells were seeded in black 96-well plates and treated with increasing concentrations of RP-6306 for 48 or 72 hours. Cell viability was assessed using the Deep Blue Cell Viability Kit and measured with a Varioskan Flash fluorometer using SkanIt version 2.4.3. Treatments were performed at a density of

infiltration was assessed at end point by flow cytometric quantification of GFP⁺ MM5080 cells, revealing a reduced frequency of tumor cells in the BM of treated mice compared with controls (Figure 5J). These data indicate that RP-6306 exerts potent antitumor activity against *Trp53*-deficient MM cells in both in vitro and in vivo settings.

Discussion

In MM, del(17p) is a high-risk cytogenetic abnormality strongly associated with poor clinical outcome, underscoring its significance as a prognostic marker and a therapeutic challenge.^{5,6} Here, we aimed to uncover potential vulnerabilities imparted by 17p loss and, in turn, identify genetic and pharmacological dependencies that could be exploited to target essential survival pathways activated by 17p deletion. Although loss of *TP53* located on 17p has been strongly associated with disruptions in the G1/S checkpoint, impaired DNA damage response, and defective repair mechanisms, ultimately driving genomic instability and tumor progression,^{24,25} available data suggest the role of additional genes, perturbed by del(17p), in driving the functional phenotype of these MM cells through various regulatory and compensatory pathways. In this study, we sought to identify genes that might drive these compensatory mechanisms, providing a functional advantage to del(17p) MM cells.

We identified PKMYT1, a G2/M checkpoint regulator, as a novel vulnerability in MM cells with del(17p) and/or *TP53* mutations. PKMYT1 is a serine/threonine kinase and a member of the WEE1 family of proteins with a significant role in the G2/M checkpoint of the cell cycle. It phosphorylates CDK1 and prevents early entry of the cell into mitosis by maintaining CDK1-cyclin B1 in an inactive state.¹¹ The early mitotic entry triggered by PKMYT1 inhibition, coupled with CDK1 hyperactivation, has been shown to be particularly damaging to certain cancer cells because it exacerbates replication stress and synergizes with DNA damage, underscoring the importance of PKMYT1 as a vulnerability in cancer cells experiencing high replication stress.^{17,26} Our findings further extend the therapeutic relevance of PKMYT1 inhibition to MM, particularly in the context of 17p deletion and *TP53* dysfunction. Interestingly, previous studies in pancreatic ductal adenocarcinoma have shown that abrogation of the G2 checkpoint via PKMYT1 inhibition is synthetically lethal in p53-deficient tumor cells,¹⁵ aligning with our observations in MM.

While *WEE1* is recognized as an essential gene, PKMYT1 stands out as a potentially more selective therapeutic target, because its inhibition does not appear to significantly affect long-term cell growth in normal cells.²⁷ *WEE1* inhibitors have been previously explored in MM,²⁸ but their clinical application has been hindered by pleiotropic effects and associated toxicities. Furthermore, PKMYT1 has been identified as a

candidate predictive biomarker for acquired resistance to *WEE1* inhibitors,²⁹ and dual *WEE1* and PKMYT1 inhibition has been proposed as a strategy to enhance tumor cell selectivity while reducing the required doses of each inhibitor, thereby minimizing toxicity.³⁰ Our study provides a fresh perspective by identifying PKMYT1 as a promising and potentially more targeted therapeutic vulnerability in MM cells with 17p loss, paving the way for innovative treatment strategies tailored to the high-risk patient population.

CCNE1 amplification and mutations in *FBXW7* or *PPP2R1A* have been associated with RP-6306 sensitivity in other cancer cell contexts.¹⁷⁻¹⁹ In our panel of MM cell lines, we observed a trend between RP-6306 response and cyclin E expression, although not significant, as well as a trend toward increased *CCNE1* expression in patients with del(17p) (data not shown). However, because of the lack of matched genomic annotations for the specific models tested, we could not evaluate whether co-occurring alterations (eg, *FBXW7* or *PPP2R1A* mutations) contribute to the increased sensitivity observed in *TP53*-altered cells. Future studies using genomically annotated cohorts and/or isogenic models will be required to determine whether RP-6306 sensitivity is driven by *TP53* alterations alone or in combination with additional genetic features.

We, however, believe that PKMYT1 is not inherently “irrelevant” in patients with *TP53* WT. Instead, we propose that PKMYT1 inhibition is most effective in cells with high baseline replication stress that rely on PKMYT1 to restrain CDK1 and prevent premature mitotic entry. In this context, *TP53*-altered cells may be more sensitive because loss of p53 weakens the G1/S checkpoint, increasing reliance on the G2/M checkpoint. As a result, PKMYT1 inhibition may have a larger therapeutic window in *TP53*-altered disease, whereas *TP53* WT cells may be less sensitive or require combination strategies.

By activating CDK1, PKMYT1 inhibition drives cells into premature mitosis with underreplicated and damaged DNA, ultimately resulting in mitotic catastrophe. In parallel experiments, we observed that cotreatment with the CDK1 inhibitor RO-3306 only partially rescued RP-6306-induced cell death in del(17p) MM cells. This indicates a potential role of other kinases in driving the observed phenotype. These intermediates may be responsible for both perturbation of PKMYT1 activity in del(17p) MM and the selective sensitivity of del(17p) MM cells to RP-6306. Further investigations will be necessary to uncover these potential contributors and fully understand the underlying mechanisms.

To further expand therapeutic possibilities, we investigated combination strategies and observed synergistic activity when PKMYT1 inhibition was combined with DNA-damaging agents. These data suggest that in del(17p) cells, which often lack functional *TP53* and therefore have impaired DNA damage

Figure 5 (continued) 0.25 × 10⁶ cells per mL, and all experiments were conducted in triplicate. Error bars represent mean ± SD. (I) Transplantation of MM5080 cells carrying *Trp53* loss was conducted in immunocompetent C57BL/6 mice by IV injection. Mice (n = 5 per group) were randomly assigned to 3 treatment groups: (1) vehicle, (2) RP-6306 at 5 mg/kg, or (3) RP-6306 at 10 mg/kg. Treatment was initiated on day +3 after injection. RP-6306 was administered orally twice daily, 5 days per week, for 3 weeks. Therapeutic responses were evaluated using a log-rank test. (J) BM infiltration in the MM5080 transplant model was assessed at end point by flow cytometric quantification of GFP⁺ MM5080 cells in the BM. Values are shown as the percentage of MM5080 cells normalized to the mean of the vehicle-treated group. Error bars represent mean ± SD. *P* values were determined using the Kruskal-Wallis test. Hb, hemoglobin; PLT, platelet; SCID, severe combined immunodeficiency; WBC, white blood cell count.

responses, PKMYT1 inhibition could further compromise the cells' ability to manage genomic stress. As a result, when exposed to DNA-damaging agents, these cells accumulate unrepaired lesions more rapidly, leading to enhanced cytotoxicity. These results underscore the potential of leveraging genomic vulnerabilities in MM cells through combination approaches.

In addition, studies in other cancers have demonstrated that RP-6306 treatment can induce interferon responses and promote T-cell infiltration,^{31,32} suggesting that PKMYT1 inhibition may enhance antitumor immune responses and increase susceptibility to immunotherapy. These findings highlight the need for further studies to explore the role of RP-6306 in modulating immune responses within the MM tumor microenvironment, as well as the potential for integrating PKMYT1 inhibition into combination strategies that include immunotherapy.

In summary, we highlight PKMYT1 as a novel and selective vulnerability specifically in del(17p) MM and its importance as a therapeutic target, offering a promising option for patients with limited treatment outcomes. In the era of risk-adapted treatment and biomarker-driven therapeutic interventions, our finding that PKMYT1 is a specific vulnerability in patients with del(17p) has the potential to be a personalized medicine approach for this high-risk entity.

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Authorship

Contribution: A.S., M.F., and N.C.M. conceived and designed the study and wrote the manuscript; A.S., S.M., and M.K.S. performed RNA sequencing; A.S., M.K.S., and A.A.-S. analyzed patient RNA sequencing data; A.S. analyzed DepMap data; J. Cui and J.E.M. performed in vitro experiments and analyzed the data with M.F. and A.S.; J. Cui and J.E.M. performed in vivo experiments using a humanized subcutaneous

MM model and analyzed the data with M.F.; C.C. analyzed patient single-cell data; S.T. and M.S. performed micronucleus assay; V.F. and A.S. performed immunofluorescence experiments; C.V., M.L., and J.A.M.-C. performed in vivo murine myeloma model experiments; D.P. performed the ExviTech automated flow cytometry analysis; J. Corre, H.A.-L., and K.C.A. critically evaluated the experimental data and manuscript; and all authors reviewed and approved the manuscript.

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Footnotes

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The deidentified and annotated processed data elements are available from the corresponding author, Mariateresa Fulciniti (Mariateresa_fulciniti@dfci.harvard.edu), on request, or at <https://www.myelomaresearch.net/faq>.

The online version of this article contains a data supplement.

There is a *Blood Commentary* on this article in this issue.

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